Teste Abrangente de Formulas Matematicas — Typst

1. Aritmetica Basica e Fracoes

$$a + b = c$$

$$a - b = c$$

$$a \times b = c$$

$$a \cdot b = c$$

$$\frac{a}{b} = c$$

$$\frac{a + b}{c + d} = e$$

$$a^2 + b^2 = c^2$$

$$\sqrt{a^2 + b^2} = c$$

$$\sqrt[3]{x} = y$$

2. Simbolos Gregos

$$\alpha + \beta = \gamma$$

$$\mathrm{A} + \mathrm{B} = \Gamma$$

$$\delta, \Delta, \varepsilon, \epsilon$$

$$\zeta, \eta, \theta, \vartheta, \iota, \kappa$$

$$\lambda, \Lambda, \mu, \nu, \xi, \Xi$$

$$\pi, \Pi, \rho, \varrho, \sigma, \Sigma, \varsigma$$

$$\tau, \upsilon, \Upsilon, \varphi, \phi, \Phi$$

$$\chi, \psi, \Psi, \omega, \Omega$$

3. Relacoes e Operadores Logicos

$$a < b$$

$$a > b$$

$$a \leq b$$

$$a \geq b$$

$$a \neq b$$

$$a \approx b$$

$$a \equiv b$$

$$a \propto b$$

$$a \in B$$

$$a \subset B$$

$$a \subseteq B$$

$$a \supset B$$

$$a \supseteq B$$

$$a \cup B$$

$$a \cap B$$

$$\varnothing$$

$$\forall x \in \mathbb{R}$$

$$\exists y \in \mathbb{N}$$

$$\neg(a = b)$$

$$a \wedge b$$

$$a \vee b$$

$$a \Rightarrow b$$

$$a \Leftrightarrow b$$

4. Calculo e Analise

$$\frac{\mathrm{d}}{\mathrm{d}x} f(x) = f'(x)$$

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2}$$

$$\int_a^b f(x) \mathrm{d}x$$

$$\int_0^\infty e^{-x} \mathrm{d}x = 1$$

$$\oint_C f(z) \mathrm{d}z = 2\pi i$$

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

$$\prod_{k=1}^{n} k = n!$$

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \to \infty} \frac{1}{x} = 0$$

$$\frac{\partial}{\partial x} f(x, y)$$

$$\nabla f(x, y, z)$$

5. Algebra Linear

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$A^T$$

$$A^{-1}$$

$$A^\dagger$$

$$\mathrm{tr}(A)$$

$$\mathrm{rank}(A)$$

$$\dim(V)$$

$$\ker(A)$$

$$\mathrm{im}(A)$$

$$\langle u, v \rangle$$

$$\|v\|$$

$$|x|$$

$$\lfloor x \rfloor$$

$$\lceil x \rceil$$

6. Funcoes Especiais

$$\sin x, \cos x, \tan x$$

$$\arcsin x, \arccos x, \arctan x$$

$$\sinh x, \cosh x, \tanh x$$

$$\log_a x, \ln x, \exp x$$

$$\max(a,\ b), \min(a,\ b)$$

$$\gcd(a,\ b), \mathrm{lcm}(a,\ b)$$

$$\lfloor x \rfloor, \lceil x \rceil, \lfloor x \rceil$$

$$\mathrm{sgn}(x)$$

$$\mathrm{erf}(x)$$

$$\mathrm{Res}(f(z), z = z_0)$$

7. Conjuntos e Teoria dos Numeros

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

$$\{x \in \mathbb{R} \mid x > 0\}$$

$$\mathcal{P}(A)$$

$$\aleph_0$$

$$\beth_1$$

$$\binom{n}{k}$$

$$n! = n \cdot (n - 1) \cdot ... \cdot 1$$

$$\begin{matrix} (n \\ k) = \frac{n!}{k!(n-k)!} \end{matrix}$$

8. Alinhamento de Equacoes

$$\sum_{k=0}^{n} k = 1 + ... + n \backslash = \frac{n(n+1)}{2} \backslash = \frac{n^2}{2} + \frac{n}{2}$$

$\frac{3x + y}{7} = 9$ dado \ $3x + y = 63$ multiplique por 7 \ $3x = 63 - y$ subtraia y \ $x = 21 - \frac{y}{3}$ divida por 3

9. Funcoes por Partes e Casos

$$f(x) = \begin{cases} 0 & \text{se } x < 0 \\ x^2 & \text{se } 0 \leq x < 1 \\ 1 & \text{se } x \geq 1 \end{cases}$$

$$\mid x \mid = \begin{cases} x & \text{se } x \geq 0 \\ -x & \text{se } x < 0 \end{cases}$$

10. Decoradores e Acentos

$$\hat{x}$$

$$\tilde{x}$$

$$|x|$$

$$(x)$$

$$\dot{x}$$

$$\ddot{x}$$

$$\underbrace{a + b + c}_{\text{soma}}$$

$$\overbrace{a + b + c}^{\text{soma}}$$

$$\underline{a + b + c}$$

$$\overline{a + b + c}$$

$$\cancel{a + b}$$

$$\sout{a + b}$$

11. Fisica e Quimica

$$E = mc^2$$

$$F = G \frac{m_1 m_2}{r^2}$$

$$p = mv$$

$$L = r \times p$$

$$H\psi = E\psi$$

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$\nabla \times B = \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial E}{\partial t}$$

$$\mathrm{d}S \geq \frac{\delta q}{T}$$

$$pV = nRT$$

$$\lambda = \frac{h}{p}$$

12. Probabilidade e Estatistica

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

$$E[X] = \sum_i x_i P(X = x_i)$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

$$X \sim N(\mu, \sigma^2)$$

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} \mathrm{d}t$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

13. Geometria e Topologia

$$\pi r^2$$

$$2\pi r$$

$$\frac{4}{3} \pi r^3$$

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

$$d(x, y) = \max_i \mid x_i - y_i \mid$$

$$d(x, y) = \sum_{i=1}^{n} \mid x_i - y_i \mid$$

$$\chi(M) = V - E + F$$

14. Numeros Complexos

$$i^2 = -1$$

$$z = a + bi$$

$$\mid z \mid = \sqrt{a^2 + b^2}$$

$$z^* = a - bi$$

$$e^{i\theta} = \cos\theta + i \sin\theta$$

$$e^{i\pi} + 1 = 0$$

15. Series e Transformadas

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} \mathrm{d}t$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} \mathrm{d}\omega$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x} \text{ para } \mid x \mid < 1$$

16. Fontes e Estilos Especiais

$$\mathbb{R}$$

$$\mathcal{L}$$

$$\mathfrak{G}$$

$$\mathtt{code}$$

$$\text{text}$$

$$\mathsf{sans}$$

$$\mathrm{x}$$

$$\boldsymbol{x}$$

$$x$$

17. Subscritos e Sobrescritos Complexos

$$x_{i,j}^{k,l}$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij}$$

$$\int_V f(x) \mathrm{d}V$$

$$\bigcup_{i \in I} A_i$$

$$\bigcap_{i \in I} A_i$$

$$\infty$$

$$\aleph$$

$$\beth$$

18. Expressoes Aninhadas

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + x}}}$$

$$\sqrt{a + \sqrt{b + \sqrt{c}}}$$

$$\frac{e^{x^2 + y^2}}{x^2 + y^2}$$

$$\frac{\partial^2 f}{\partial x \partial y}$$

$$\overbrace{\underbrace{a + b}_{\text{bottom}}}^{\text{top}}$$

19. Espacamento e Tamanhos

$$a\,b$$

$$a\;b$$

$$a\ b$$

$$a\quad b$$

$$a\qquad b$$

$$\frac{1}{2}$$

$$\dfrac{1}{2}$$

$$\tfrac{1}{2}$$

$$\scriptstyle \frac{1}{2}$$

20. Referencias Cruzadas e Numeracao

$$E = mc^2 \quad (1)$$

$$F = ma \quad (2)$$

Referindo-se as equacoes Equation 1 e Equation 2.

21. Matrizes Avancadas

$$\begin{Bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{Bmatrix} \quad (3)$$

$$\begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{pmatrix} \quad (4)$$

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ 2 & 3 & \cdots & n+1 \\ \vdots & \vdots & \ddots & \vdots \\ n & n+1 & \cdots & 2n-1 \end{pmatrix} \quad (5)$$

22. Delimitadores Escalaveis

$$\left(\frac{a}{b}\right) \quad (6)$$

$$\left[\frac{a}{b}\right] \quad (7)$$

$$\left\{\frac{a}{b}\right\} \quad (8)$$

$$\left|\frac{a}{b}\right| \quad (9)$$

$$\left\lfloor\frac{a}{b}\right\rfloor \quad (10)$$

$$\left\lceil\frac{a}{b}\right\rceil \quad (11)$$

$$\left\langle\frac{a}{b}\right\rangle \quad (12)$$

$$] \frac{a}{b} [\quad (13)$$

23. Operadores Personalizados

$$\mathrm{Res}(f(z),\ z = z_0) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z - z_0} \mathrm{d}z \quad (14)$$

$$\mathrm{Var}(X) = E\left[(X - \mu)^2\right] \quad (15)$$

$$\mathrm{Cov}(X,\ Y) = E[(X - \mu_X)(Y - \mu_Y)] \quad (16)$$

24. Equacoes com Texto

$$Q = \rho A v + \text{time offset} \quad (17)$$

$$\mathcal{A} := \{x \in \mathbb{R} \mid x \text{ is natural}\} \quad (18)$$

$$f(x) = x^2 \text{ for all } x \in \mathbb{R} \quad (19)$$

25. Analise Complexa

$$\oint_{|z|=R} f(z) \mathrm{d}z = 2\pi i \sum_{k=1}^{n} \mathrm{Res}\ (f, a_k) \quad (20)$$

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n \quad (21)$$

$$\mathrm{Li}_2(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^2} \quad (22)$$

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} \mathrm{d}t \quad (23)$$

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{prime}} \frac{1}{1 - p^{-s}} \quad (24)$$

26. Mecanica Quantica

$$[\hat{x}, \hat{p}] = i\ \hbar \quad (25)$$

$$i\ \hbar\ \frac{\partial \Psi}{\partial t} = \hat{H} \Psi \quad (26)$$

$$\Psi(x,\ t) = \sum_n c_n \psi_{n(x)} e^{-i E_n \frac{t}{\hbar}} \quad (27)$$

$$\langle \varphi || \psi \rangle = \int \varphi^*(x) \psi(x) \mathrm{d}x \quad (28)$$

$$\langle A\hat{} \rangle = \langle \psi | \hat{A} | \psi \rangle \quad (29)$$

27. Relatividade

$$\mathrm{d}s^2 = -c^2 \mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \quad (30)$$

$$\Gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (31)$$

$$E^2 = (pc)^2 + (mc^2)^2 \quad (32)$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (33)$$

28. Teoria dos Grafos

$$\chi(G) \geq \omega(G) \quad (34)$$

$$\alpha(G) + \tau(G) = \mid V(G) \mid \quad (35)$$

$$\sum_{v \in V(G)} \deg(v) = 2 \mid E(G) \mid \quad (36)$$

$$K_n, C_n, P_n, Q_n \quad (37)$$

29. Teoria da Informacao

$$H(X) = -\sum_{x \in X} P(x) \log_2 P(x) \quad (38)$$

$$I(X; Y) = H(X) - H(X \mid Y) \quad (39)$$

$$D_{KL}(P \mid Q) = \sum_x P(x) \log\left(\frac{P(x)}{Q(x)}\right) \quad (40)$$

$$C = \max_{P_X} I(X; Y) \quad (41)$$

30. Otimizacao

$$\min_{x \in \mathbb{R}^n} f(x) \text{ sujeito a } g_{i(x)} \leq 0, h_{j(x)} = 0 \quad (42)$$

$$L(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_{i(x)} + \sum_j \mu_j h_{j(x)} \quad (43)$$

$$\nabla f(x^*) + \sum_i \lambda_i^* \nabla g_{i(x^*)} + \sum_j \mu_j^* \nabla h_{j(x^*)} = 0 \quad (44)$$